

Selected Topics in AI: Final Project

Part 1: LLM powered Chatbot using Streamlit with RAG

1. Use Ollama for choosing open-source LLMs for question-answering and summarizing files such as pdf, csv and arxiv papers.
2. Your chatbot should support the previously mentioned types of documents for question answering and summarization.
3. Your chatbot should support memory.
4. You should show the whole conversation between the user and the chatbot.

Part 2 : Multimodal langchain

1. Choose a multimodal task combining NLP and Computer Vision such as image captioning or visual question answering or any other task of your choice.
2. Choose a transformer-based model for the chosen task.
3. Augment your previously created chatbot with computer vision task support.

Bonus: Audio I/O

1. Add an option for audio I/O, reading the generated responses aloud and the chatbot accepts audio input for the prompt.

Notes:

- You should work in teams of 2. Working individually is not allowed.

Rubric:

Part 1.1, Part 1.2	50%
Part 1.3	15%
Part 1.5	15%
Part 2	20%